

TED UNIVERSITY

DEPARTMENT OF COMPUTER ENGINEERING

SENIOR DESIGN PROJECT PROPOSAL

1. Project & Committee Information

Project Name	ERDENIS: Reinforcement Learning Based Attack and Evasion Decision-Making for UAVs
Project Web Page URL	https://erdenis.com.tr
Project Supervisor	Prof. Dr. Gökçe NUR YILMAZ
Jury Members	1. Dr. Fırat Akba 2. Prof. Dr. Tolga Kurtuluş Çapın

2. Team Members

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3. Project Description

This project focuses on the development of a decision-making framework using reinforcement learning that will allow the unmanned aerial vehicles to perform attack and evasion maneuvers in the case of an advantageous situation or any danger posed to them in the course of their flight. This will be accomplished through the use of a simulation tools like AirSim or Gazebo and training an agent to choose proper avoidance maneuvers while staying within the confines of the vehicle's capabilities.

The developed decision-making structure will be compared with currently available approaches and evaluated based on certain performance criteria such as rate of successful attack, avoidance and collision across the scenarios that have been simulated. The expected output of the project will include the creation of simulation scenarios, decision-making framework and comparison of the learning-based and conventional evasion scheme.